



B.Tech. Project (BTP) / Industrial Training Project (ITP) Guidelines

Preamble

1. A student is required to take at least a semester long BTech project (BTP), during which he/she is required to demonstrate his/her ability to learn current areas of research and/or industrial interest.
2. There is a distinction between the on-campus project (BTP) and the off-campus project, called as Industrial Training Project (ITP).
3. A student in the program can choose to divide his/her BTP work between 7th and 8th semester by taking 3 credits (0-1-6) of BTP Part 1 in 7th semester and 9 credits (0-2-12) of BTP Part 2 in 8th semester. The student can also do his/her BTP work entirely in semester 8.
4. Instead of a BTP, a student has an option to do an Industrial Training Project (ITP) in the 8th semester only and in the off-campus mode.
5. ITP is to be done at an approved industry site with a designated industry mentor on a well specified project. This option is suitable for those students who want to gain an early industry experience to better prepare them as an industry professional. This option may also be suitable for students who wish to gain this experience before heading out for higher studies.
6. In the BTech project (BTP) the student undertakes a problem of interest, identifies the issues involved, and develops techniques to address the issues. The student can undertake a theoretical study and/or experimental or developmental work. The work can be carried out individually or in groups under the supervision of faculty. The BTP work can be a continuation of the work done during previous internships.

Guidelines:

1. BTP/ITP should generally commence by the first week of January and last till the last week of April. It should last for a minimum of 13 weeks (continuous) before a student can appear for the BTP/ITP presentation.
2. A group of not more than 2 students can work on a common problem for a BTP under a faculty mentor.

3. A member of faculty is recommended to supervise not more than 6 BTPs, with a total strength of not more than 10 students.
4. A student can get to do an ITP by participating and qualifying for a company in the internship placement drive conducted by the placement office, DAIICT.
5. Students can also do ITPs in companies through their own contacts. But in such a case, not more than 10 students will be permitted to intern in a company/organization at a time, unless the company formally approaches the placement office for recruiting students through the internship placement drive of DAIICT. Self-arrangement ITPs is encouraged in R&D establishments.

ITP Evaluation:

1. For Part 1, the on campus mentor communicates with the onsite mentor during middle of the Winter semester and receives a grade from them, after possibly some discussions. This is then uploaded into the USPMES.
2. For Part 2, the students need to send a report to the on campus mentor. This report is put through a plagiarism check using the software Turnitin. This is used to screen reports crossing a threshold (around 25% cumulative and 10% single source). Apart from this, the on campus mentor again receives a grade from the onsite mentor and uploads this on the USPMES system.
3. An evaluation committee is formed at the end of the semester by the BTP/ITP Coordinator to whom the students returning from their off campus projects make a presentation. On the basis of the presentation and the report, the committee either by consensus or individually gives student a grade.
4. The scores of the committee are collected by the BTP coordinator in the form of a hard copy. The committee also has the on campus mentor who also participates in the grading process
5. The final grade of the student for Part 2 is calculated giving 50% weightage to the grade received from the onsite mentor and 50% weightage to the grade awarded by the committee.
6. The grade for Part 1 is uploaded by the BTP coordinator from the USPMES to the e campus. For Part 2, the BTP coordinator uploads the grade on e-campus based on the calculation of the previous step.

BTP Evaluation:

1. For Part 1, the mentor decides the grade of the student based on their interactions and assessment. This is uploaded on USPMES towards the end of Autumn semester if the BTP part 1 is done in the Autumn semester, otherwise, in the middle of the winter semester.
2. For Part 2, the mentor asks the student to write a report, which is also checked for plagiarism using Turnitin, apart from having the mentor's approval. The mentor uploads a

- grade as per their assessment of the student on e-campus. This has 70% weightage for the final grade the student receives for Part 2.
3. An evaluation committee is formed with the mentor being a part of it. However, the grade assigned by the committee either by consensus or individually does not involve the mentor. This grade is received in the form of a hard copy by the BTP coordinator and has 30% weightage for the final grade of Part 2.
 4. The BTP coordinator uploads the grade for Part 1 on e-campus as it appears on USPMES. For Part 2, the grade is uploaded on e-campus on the basis of the calculation in the previous step.

Acceptable BTPs/ITPs:

BTP/ITP can be on a novel/new idea, an extension of some previous research work, an extension of some previous BTP, an abstract problem or some proof of concept problem. The possible areas of work are:

- Software Development
- System Design and Simulation
- Hardware Development/Implementation
- Embedded System (Software & Hardware combine) Development/Implementation
- Theoretical Modeling
- Technical Study
- Technical Survey and Modeling
- Modules of a big research and development project jointly guided by teams of faculty with a focus on synthesis of their class-room learning to solve real world problems

Deliverable of a BTP/ITP:

In general, the expected deliverables will include, among others, reporting of:

- Problem statement
- Methodology adopted
- Background work
- Comparative study (products/methods/designs)
- Design documentation
- Simulation study results
- Analysis results
- Test methods and results
- Conclusive contribution of the project

In case certain projects may not get clearly classified amongst the categories mentioned above, the concerned supervisor may identify the specific deliverables for those projects.